Technical review

Item #4 · signed-then-modified.pdf · Run ID: 20261113T232756Z-94fc69

Path	`R:\evidence\pdf\signed-then-modified.pdf`
Size (bytes)	11980
Last modified (UTC)	2026-11-12T06:00:00Z
Extension	.pdf
Detected type (signature)	PDF document
Status	Problem found
Case	RV-SAMPLE-001
Evidence ID	E0021
Case registry	Verified: identical to the registered evidence
Registered at intake (UTC)	2026-11-13T23:27:52.795Z

Hashes

sha256	`a6e9f4503e7a5fd6fabfff5a2e7b99d949f934e04a65dfe46814cc8ed578a0b3`
md5	`6f259789b37be1219f161c23960fd169`

Findings

[High] Content changed after signing (CONTENT_CHANGED_AFTER_SIGNATURE)

What it shows	Signature Signature1 covers an earlier revision; revision(s) 3, added afterwards, change page content or document structure, which the signature does not cover.
What it does not prove	The signature remains mathematically valid for the signed revision; this does not show who made the later changes or whether the signer agreed to them. Compare the signed revision with the final one.

[High] Changes not allowed by the certification signature (DOCMDP_VIOLATION)

What it shows	Signature Signature1 certifies the document with permission level 2 (1 = no changes, 2 = form filling and signing, 3 = also annotations), but revision(s) 3 contain changes outside that level.
What it does not prove	The classification of changes is heuristic and some PDF viewers apply permissions differently; this shows a conflict with the declared permissions, not intent.

[Low] Incremental updates (INCREMENTAL_UPDATES)

What it shows	The file contains 3 revisions: the original save plus later updates appended to the end of the file. Each earlier version can be extracted and compared.
What it does not prove	Incremental saving is normal when a PDF is filled in, signed, annotated or re-saved by many programs; it does not show who changed the file or that a change was improper.

[Informational] Time gap between revisions (REVISION_TIME_GAP)

What it shows	Revision 3 is dated 3 d 0 h 0 min after revision 2 (according to their modification dates).
What it does not prove	Metadata dates are written by the editing software from the computer clock and can be set freely; the gap does not prove when the changes were really made or by whom.

[Informational] Digital signature present (trust not verified) (SIGNATURE_PRESENT_TRUST_NOT_VERIFIED)

What it shows	The file contains signature Signature1, signed by O=Rakvor TEST,CN=Ana Ruiz (TEST examiner). Its byte coverage and cryptographic integrity are reported separately.
What it does not prove	This tool does not validate the certificate chain, revocation or the signer's identity; a signature can be technically intact and still come from an untrusted or self-made certificate.

[Medium] Visible change between revisions (VISUAL_CHANGE_BETWEEN_REVISIONS)

What it shows	Rendering revision 2 and revision 3 gives different images on page(s) 1 (largest changed share 0.000314); pages added: -; pages removed: -. The changed areas are outlined in the comparison images.
What it does not prove	Incremental updates are normal when forms are filled, documents are signed or comments are added. It does not show who made the change, when, or with what intent.

Revision history

#	Byte range	Saved at (Info)	Saved at (XMP)	Producer	Time since previous	Objects added / changed / deleted	Pages changed	Signed by (field)
1	0–1646 (1646 B)	2026-05-04T13:00:00Z	—	ReportLab PDF Library - (opensource)	—	8 / 0 / 0	—	—
2	1646–10975 (9329 B)	2026-05-04T13:00:00Z	—	ReportLab PDF Library - (opensource)	less than a minute	3 / 2 / 0	—	Signature1
3	10975–11980 (1005 B)	2026-05-07T13:00:00Z	—	ReportLab PDF Library - (opensource)	3 days	1 / 2 / 0	1	—

Timeline

#	Saved at (UTC)	Time since previous	Producer
1	2026-05-04 13:00:00	—	ReportLab PDF Library - (opensource)
2	2026-05-04 13:00:00	less than a minute	ReportLab PDF Library - (opensource)
3	2026-05-07 13:00:00	3 days	ReportLab PDF Library - (opensource)

Dates are those stored in the file (Info dictionary, else XMP); the software that saved it can set them freely. Without usable dates the revisions are spaced evenly.

Signatures

Field	Signer (certificate subject)	Signing time (claimed)	Revision	Covers revision	Digest matches	Signature value	DocMDP	Later revisions
Signature1	O=Rakvor TEST,CN=Ana Ruiz (TEST examiner)	2026-05-04T13:20:00Z	2	yes	yes	yes	2	3

The signer certificate, its chain and its revocation status were not validated: a mathematically valid signature does not show who controlled the key.

Fonts

Font	Subset prefix	Embedded	First revision	Object
Helvetica	—	no	1	2 0
Helvetica-Bold	—	no	1	3 0

Producer fingerprints

Role	Database	Version	Entries	Status	SHA-256
starter	rakvor-starter	2026.10.21	37	loaded	`0b8df17b3ba7f33932cfc072932f198cdfdba98ad1332b33ac84c5c2f9068795`
local	—	—	—	absent	—

#	Tool	Category	Matched on	Value found	Traits not matching
1	ReportLab PDF Library	pdf_library	producer	ReportLab PDF Library - (opensource)	—

Visual comparison

Status	Compared
Pages changed	1

Difference images (in this item's folder):

- visual/rev-02-vs-03-page-001.png

Per-page measurements

Revisions	Page	Changed	Changed share	Changed pixels	Regions	DPI	Page size changed	Image / reason
1?2	1	no	0.0000 %	0	0	110	no	—
2?3	1	yes	0.0314 %	356	2	110	no	visual/rev-02-vs-03-page-001.png

Parser warnings

None

The complete list of parser warnings is in warnings.json in this item's folder (its digest is in the audit log and in the extracted files table).

Extracted files

warnings.json (parser_warnings, 123 B)	md5: 251720985f0ba1c888116538bc3da921 sha256: 9dfdca46175986eda2dbcd6912894e5d123f4f0d267ee0e0b018d8d96fc149ae
revisions/rev-01.pdf (revision, 1646 B)	md5: 1fcc3fbbcdfeb78ad30a7097b7c7f9c2 sha256: 63cbba08c3f6ac14827c95cf554a0fb5180dbe9f6cfc11d399c43e4401deb94a
revisions/rev-02.pdf (revision, 10975 B)	md5: 2003920f3ccc73ba80d8677562e97e5b sha256: 9bcad4dfc32943d7b047f5ce98e37e5bee63f580ab9e547526495fcec1851dbf

revisions/rev-03.pdf (revision, 11980 B)	md5: 6f259789b37be1219f161c23960fd169 sha256: a6e9f4503e7a5fd6fabfff5a2e7b99d949f934e04a65dfe46814cc8ed578a0b3
objects.json (objects_dump, 7550 B)	md5: 7d044db38918abc53413a8310cbbe0e3 sha256: 8f61a943df1fecbc8875b2151044a819785bcddd8934fcca944a499491430843
visual/rev-02-vs-03-page-001.png (visual_diff, 29281 B)	md5: 877c0147b627af6d6766bd5ead608012 sha256: f60dae90cb2300301de4ae82bdb132847cf40480b60d56b37f64423495f062cc

Method

Each file was opened read-only and hashed in a single pass with the selected algorithms. PDF files were then parsed without being modified: every saved revision, its dates and producer, digital signatures, fonts and producer fingerprints were recorded, and pages of consecutive revisions were compared visually. Every action was written to a hash-chained audit log.

Audit trail

Audit entries	#36 – #46
Log folder	`R:\case\runs\rv-pdfauth\20261113T232756Z-94fc69\logs`
Digital signature	This PDF is not digitally signed.

Limitations

- Type detection uses a limited table of common file signatures; unknown types are reported as unknown, not as suspicious.
- Modification times come from the file system of the examined copy and may differ from the original source.
- A result without findings does not prove that a file is authentic or unaltered before acquisition.
- A trusted timestamp proves the audit log existed in this exact form at the stated time; it does not prove that the examined copy matches the original source.
- The absence of findings does not prove that a PDF is authentic or unaltered: a document can be forged before it is saved, or re-created so that no trace remains.
- When a PDF is printed to PDF, flattened, optimised or regenerated, its earlier revisions and most editing traces are lost; only its last state can be examined.
- Producer fingerprints come from a limited database (vendor documentation and files generated by Rakvor); producer strings can be typed by hand, and unknown or spoofed values cannot be recognised.
- Digital signatures are checked for byte coverage and cryptographic integrity only; certificate chains, revocation and the signer's identity are not validated.
- The visual comparison between revisions renders only a limited number of pages per revision; pages beyond that limit are compared structurally only.